

Regular Expressions in Programming Languages: Java for You

This is the sixth and final part of a series of articles on regular expressions in programming languages. In this article, we will discuss the use of regular expressions in Java, a very powerful programming language.



Java is an object-oriented general-purpose programming language. Java applications are initially compiled to bytecode, which can then be run on a Java virtual machine (JVM), independent of the underlying computer architecture. According to Wikipedia, “A Java virtual machine is an abstract computing machine that enables a computer to run a Java program.” Don’t get confused with this complicated definition—just imagine that JVM acts as software capable of running Java bytecode. JVM acts as an interpreter for Java bytecode. This is the reason why Java is often called a compiled and interpreted language. The development of Java—initially called Oak—began in 1991 by James Gosling, Mike Sheridan and Patrick Naughton. The first public implementation of Java was released as Java 1.0 in 1996 by Sun Microsystems. Currently, Oracle Corporation owns Sun Microsystems. Unlike many other programming languages, Java has a mascot called Duke (shown in Figure 1).

As with previous articles in this series I really wanted to begin with a brief discussion

about the history of Java by describing the different platforms and versions of Java. But here I am at a loss. The availability of a large number of Java platforms and the complicated version numbering scheme followed by Sun Microsystems makes such a discussion difficult. For example, in order to explain terms like Java 2, Java SE, Core Java, JDK, Java EE, etc, in detail, a series of articles might be required. Such a discussion about the history of Java might be a worthy pursuit for another time but definitely not for this article. So, all I am going to do is explain a few key points regarding various Java implementations.

First of all, Java Card, Java ME (Micro Edition), Java SE (Standard Edition) and Java EE (Enterprise Edition) are all different Java platforms that target different classes of devices and application domains. For example, Java SE is customised for general-purpose use on desktop PCs, servers and similar devices. Another important question that requires an answer is, ‘What is the difference between Java SE and Java 2?’ Books like ‘Learn Java 2 in 48 Hours’ or ‘Learn Java SE in Two Days’ can confuse beginners a lot while making a choice. In a nutshell, there is no difference between the two. All this confusion arises due to the complicated naming convention followed by Sun Microsystems.



Figure 1: Duke – the mascot of Java